

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Roll No.: A24126552258

Date: -0-2026

1.TITLE

Design Static Webpage using HTML Components

2. AIM

To design and create a static webpage incorporating fundamental HTML elements such as headings, paragraphs, lists, tables, forms, images, and hyperlinks.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the core structure and syntax of an HTML document.
 - To use formatting elements such as headings, paragraphs, and break tags.
 - To implement ordered and unordered lists to present structured data.
 - To construct tables for tabular data layout.
 - To create an HTML form with various input controls (text, email, radio, checkbox, dropdown, button).
 - To embed multimedia elements like images and hyperlink external/internal pages.
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4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to structure web pages and their content. It consists of a series of elements represented by tags that tell the browser how to display content.

- Document Structure: Every standard HTML5 document begins with `<!DOCTYPE html>` followed by `<html>`, `<head>`, and `<body>` tags.
- Text Formatting: Tags like `<h1>` to `<h6>` define headings in order of importance, while `<p>` defines block-level paragraphs.
- Lists: `<ol>` (Ordered List) creates numbered sequences, and `<ul>` (Unordered List) creates bulleted lists using `<li>` elements.
- Tables: `<table>`, `<tr>` (table row), `<th>` (table header), and `<td>` (table data) arrange data into grid columns and rows.

- Forms: <form> collects user inputs using controls like <input>, <select>, and <textarea>.
 - Hyperlinks & Media: <a> anchors link to other pages using the href attribute, while embeds visual graphics via the src attribute.
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5. ALGORITHM / PROCEDURE

1. Open a code editor (such as VS Code or Notepad).
 2. Create a new file as static_webpage.html.
 3. Write the basic HTML5 structure (<!DOCTYPE html>, <html>, <head>, <body>).
 4. Add a title tag in the head section to name the web document.
 5. Create a header section containing main page headings (<h1>, <h2>) and horizontal lines (<hr>).
 6. Build a navigation section using hyperlink tags (<a>).
 7. Insert a content section with paragraphs, bold text, and italic text.
 8. Create an unordered list () and an ordered list () to categorize information.
 9. Construct a data table (<table>) with proper header (<th>) and data (<td>) rows.
 10. Build an input form (<form>) containing text inputs, email inputs, radio buttons, checkboxes, a drop-down list (<datalist>), and a submit button.
 11. Embed an image element () with an alternate text attribute (alt).
 12. Save the document and open static_webpage.html in any web browser.
 13. Verify that all components render correctly according to standard HTML rendering rules.
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6. PROGRAM

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Static Webpage </title>

</head>

<body>

    <!-- Header Section -->

    <header>

        <table>
```

```
<tr>

    <td>

        

    </td>

    <td>

        <h1>Department of CSM(Artificial Intelligence & Machine Learning)</h1>

    </td>

</tr>

</table>

<h2>Student-Registration portal</h2>

<hr>

<!-- Navigation Links -->

<nav>

    <a href="#about">About</a> |

    <a href="#courses">Courses</a> |

    <a href="#register">Register</a>

</nav>

</header>

<hr>

<!-- About Section -->

<section id="about">

    <h3>About the Portal</h3>

    <p>This is a <b>static webpage</b> designed using fundamental <i>HTML components</i> such
as headings, tables, forms, and lists.

    Here we can fill the students details but it is only Static.</p>

</section>

<!-- Courses List Section -->

<section id="courses">

    <h3>Available Courses</h3>

    <h4>Front-End Modules (Unordered List)</h4>

    <ul>

        <li>HTML5 Structure</li>
```

```
<li>CSS3 Styling</li>
<li>JavaScript Fundamentals</li>
</ul>

<h4>Learning Steps (Ordered List)</h4>
<ol>
    <li>Learn basic syntax</li>
    <li>Build static layouts</li>
    <li>Integrate interactive scripts</li>
</ol>
</section>

<!-- Student Data Table -->
<section>
    <h3>Enrolled Student Details</h3>
    <table border="1" cellpadding="8" cellspacing="0">
        <thead>
            <tr>
                <th>S.No</th>
                <th>Roll Number</th>
                <th>Name</th>
                <th>Branch</th>
            </tr>
        </thead>
        <tbody>
            <tr>
                <td>1</td>
                <td>A24126552258</td>
                <td>Priya Varma</td>
                <td>CSE</td>
            </tr>
            <tr>
                <td>2</td>
                <td>A24126552260</td>
```

```
        <td>Sirisha Reddy</td>
        <td>CSM</td>
    </tr>
</tbody>
</table>
</section>

<br>

<!-- Registration Form Section -->
<section id="register">
    <h3>Student Registration Form</h3>
    <form action="#" method="get">

        <label for="fullname">Full Name:</label><br>
        <input type="text" id="fullname" name="fullname" placeholder="Enter your full name"
required>

        <br><br>

        <label for="email">Email Address:</label><br>
        <input type="email" id="email" name="email" placeholder="name@example.com" required>
        <br><br>

        <label>Gender:</label><br>
        <input type="radio" id="male" name="gender" value="male">
        <label for="male">Male</label>
        <input type="radio" id="female" name="gender" value="female">
        <label for="female">Female</label>
        <br><br>

        <label for="academic-year">Academic Year:</label><br>
        <input type="text" id="academic-year" name="academic-year" list="academic-years">
        <datalist id="academic-years">
            <option value="1st year">
```

<option value="2nd year">

<option value="3rd year">

<option value="4th year">

</datalist>

<label>Interests:</label>

<input type="checkbox" id="web" name="interest" value="web">

<label for="web">Web Development</label>

<input type="checkbox" id="ai" name="interest" value="ai">

<label for="ai">ML</label>

<button type="submit">Submit Form</button>

<button type="reset">Clear Form</button>

</form>

</section>

<hr>

<footer>

<p>© 2026 Full Stack Web Development Laboratory. All rights reserved.</p>

</footer>

</body>

</html>

7. CODE EXPLANATION

Code Block / Tag	Explanation
<!DOCTYPE html>	Specifies the document type and version (HTML5).
<html>	Root element of the HTML document containing all page tags.
<head>	Holds metadata, page title, and external resource linkages.
<h1> to <h4>	Structural heading elements defining different levels of hierarchy.
<nav> & <a>	Creates structural navigational block containing target hyperlink anchors.
&	Defines unordered (bulleted) and ordered (numbered) list wrappers.
	Defines individual items inside an ordered or unordered list.
<table>, <tr>, <th>, <td>	Defines grid structures, rows, header cells, and standard data cells.
<form>	Encloses interactive user input controls for data submission.
<input>	Creates dynamic input fields based on the specified type attribute.
<datalist> & <option>	Constructs expandable drop-down selection menus.
	Embeds images specifying source file path (src) and alternative text (alt).



8. TEST CASES AND EXECUTION DATA

Test Case	Feature Tested	Input Action	Expected Result	Actual Result	Status
TC-01	Navigation Links	Click on "Register" link	Page smoothly scrolls/navigates to the Registration section	Navigated to #register target	Pass
TC-02	Text Validation	Leave "Full Name" empty & click Submit	Browser displays a validation popup requiring the field	Popup shown: "Please fill out this field"	Pass
TC-03	Radio Grouping	Select "Male" then select "Female"	Radio button selection toggles (only one remains selected)	Single selection maintained	Pass
TC-04	Form Reset	Enter sample text and click "Clear Form"	All input fields clear back to default empty states	Form cleared successfully	Pass
TC-05	Image Loading	Load webpage on browser	Image element renders inline using specified width/height	Image rendered properly	Pass

9. OUTPUT

Output 1: Header, Navigation , About Section & List Elements

Experiment 1.pdfStatic Webpage

dept of ai/ml logo - Searchdept of ai/ml logo - Search Image

127.0.0.1:5500/record/week%201-2/source%20code/static%20webpage.html?fullName=Somya+Daggubilli&email=somyadaggubilli29%40gmail.com&gender=female&academic-year=...

Chat



Department of CSM(Artificial Intelligence & Machine Learning)

Student-Registration portal

[About](#) | [Courses](#) | [Register](#)

About the Portal

This is a **static webpage** designed using fundamental *HTML components* such as headings, tables, forms, and lists. Here we can fill the students details but it is only Static.

Available Courses

Front-End Modules (Unordered List)

- HTML5 Structure
- CSS3 Styling
- JavaScript Fundamentals

Learning Steps (Ordered List)

- Learn basic syntax
- Build static layouts
- Integrate interactive scripts

Enrolled Student Details

Output 2: Enrolled Students Data Table, Student Registration Form Layout & Footer

Experiment 1.pdfStatic Webpage

dept of ai/ml logo - Searchdept of ai/ml logo - Search Image

127.0.0.1:5500/record/week%201-2/source%20code/static%20webpage.html?fullName=Somya+Daggubilli&email=somyadaggubilli29%40gmail.com&gender=female&academic-year=...

Chat

Enrolled Student Details

S.No	Roll Number	Name	Branch
1	A24126552258	Priya Varma	CSE
2	A24126552260	Sirisha Reddy	CSM

Student Registration Form

Full Name:

Email Address:

Gender:
☐ Male ☐ Female

Academic Year:

Interests:
☐ Web Development ☐ ML

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10. RESULT

Thus, a static webpage incorporating fundamental HTML components including headings, paragraphs, formatted text, lists, tables, images, hyperlinks, and forms was successfully designed, executed, and validated across multiple browser test cases.